

Sub-THz atmospheric sensing

What worked, how we achieved it, and what remains unresolved

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Problem statement

Can a satellite communication downlink also detect gas and particle concentration changes using the same transmitted signal?

The proposal covers 60–400 GHz. Gas rotational lines provide spectral structure. Small particles produce a much smoother extinction signature.

The intended outcome combines useful recall, defensible concentration limits and an acceptable communication cost.

$$\underbrace{\mathbf{a}}_{\text{observation}} = \underbrace{\mathbf{D}\mathbf{c}}_{\text{pollutants}} + \underbrace{\mathbf{N}\boldsymbol{\beta}}_{\text{background and instrument}} + \underbrace{\mathbf{e}}_{\text{noise}}$$

Parameter / check	Value
Frequency scope	60–400 GHz
Satellite altitude	550 km
Current outputs	5 VOCs + 3 PM
Family false alarms	1% budget

The practical question is whether pollutant information survives the other unknowns.

A correct forward model alone does not establish a useful environmental sensor.

Notation and units

$\mathbf{a}$: measured attenuation vector (dB)
 $\mathbf{D}$: target spectral response matrix
 $\mathbf{c}$: target concentrations ($\mu\text{g}/\text{m}^3$)

$\mathbf{N}$: background / instrument response matrix
 $\boldsymbol{\beta}$: nuisance parameter vector
 $\mathbf{D}\mathbf{c}$ and $\mathbf{N}\boldsymbol{\beta}$: attenuation contributions (dB)

$\mathbf{e}$: measurement error vector (dB)
Bold letters: vectors or matrices
Matrix products sum spectral contributions

What we have achieved

A completed computational assessment with a conditional VOC improvement

Achievement	Evidence	Boundary
Physical model	Layered gas, particle, ray and thermal calculations	Conditional atmospheric and material assumptions
VOC detection	CH ₃ CN recall 97.67% at 1 µg/m ³	20 s total and assumed 0.0001 dB residual
Engineering comparison	Sequential bands and moving coded-frame controls	RF specifications remain unmeasured
Failure diagnosis	PM information, gain drift and communication cost quantified	Useful joint VOC/PM retrieval remains open

CH₃CN interval
97.36–97.96%

Family false alarms
0.79%

Band-selection gain
40.27 percentage points

The 97.67% result uses the selected five-gas design with matched reference weather, 45° elevation, 23 dBm and 6 dB noise figure.

The original eleven tasks

Task	Original requirement	Current outcome
1.1	Stratified atmosphere	Implemented with standard and measured weather
1.2	Spectroscopic inputs	Five target VOCs and traceable catalog gaps
1.3	Line shape and slant integration	Voigt profiles and refracted spherical paths
1.4	Particle extinction baseline	Rayleigh and Mie models, negative retrieval finding
2.1	Link and radio parameters	Explicit sequential receiver and resource budget
2.2	Simulated CSI	AWGN and bounded moving waveform controls
3.1	Spectral separation	Joint inference with nuisance projection
3.2	Concentration inversion	Payload moments and signed statistical estimates
4.1	Variance lower bounds	Nuisance-aware likelihood and Fisher bounds
4.2	Sensitivity	Common VOC/PM multivariable comparisons
4.3	Detection floors and standards	Conditional limits, no positive compliance claim

Physical subtasks

4

Radio subtasks

2

Inversion subtasks

2

Performance subtasks

3

Evidence and operating conditions

Evidence	What it supports	What it cannot establish
HITRAN / ITU models	External physical parameters	Measured atmospheric radio performance
NOAA weather soundings	Actual weather profiles at selected times	Paired VOC/PM concentration truth
Monte Carlo responses	Conditional recall and concentration errors	Field recall under unknown model error
Raw coded OFDM frames	Bounded modem and synchronization checks	Full mission or continuous trajectory validation
Public water / calcite data	Limited independent physical checks	Calibrated joint VOC and PM retrieval

Trials per class

10,000

Current MC cases

12

Reference / sample

10 s / 10 s at 20 s

Measured field recall

Unavailable

Every favorable result needs its time, calibration, concentration and receiver assumptions.

Task 1.1 · Stratified atmosphere

Implemented ITU-R P.835 reference layers through 100 km and NOAA IGRA measured weather profiles.

Interpolated temperature and log pressure / water vapor across height. Explicitly modeled the continuation above each sounding.

$$n_s(z) = \frac{c_{s,0} N_A}{M_s} \exp\left(-\frac{z}{H_s}\right),$$

local spectrum depends on $P(z)$, $T(z)$, $\rho_{\text{H}_2\text{O}}(z)$

n_s is molecular number density. Use consistent mass units for $c_{s,0}$ and molar mass M_s . H_s is an assumed concentration scale height.

Parameter / check	Value
Soundings parsed	521
Vertical levels	103,322
Gas scale height	1,500 m
PM scale height	1,000 m

The channel responds to vertical weather variation instead of using one homogeneous slab.

The assumed pollutant profile does not become measured truth because the weather profile is measured.

Notation and units

s: gas species index

z: altitude (m)

$n_s(z)$: molecular number density (m^{-3})

$c_{s,0}$: surface mass concentration (kg/m^3)

N_A : Avogadro constant (mol^{-1})

M_s : gas molar mass (kg/mol)

H_s : assumed concentration scale height (m)

exp: exponential function

$P(z)$: atmospheric pressure (Pa)

$T(z)$: atmospheric temperature (K)

$\rho_{\text{H}_2\text{O}}(z)$: water vapour mass density (kg/m^3)

Convert $\mu\text{g}/\text{m}^3$ to kg/m^3 before substitution

Task 1.1 · Results and profile uncertainty

Component	Implemented result	Remaining uncertainty
Weather archive	103,322 levels from 521 soundings	Only selected dates enter sensing comparisons
Seasonal cases	January, April, July and September examples	Four profiles do not describe a population
Measured profile tops	28.8–37.0 km above the station	Upper continuation follows a model
Gas / PM scale height	1,500 m / 1,000 m	Unknown real pollutant distributions

$$c(z) = c_0 \exp\left(-\frac{z}{H}\right) \quad \Rightarrow \quad \hat{c}_0 \text{ depends on the assumed } H$$

The reported concentration is a surface-equivalent enhancement under the chosen profile.

Atmosphere top
100 km

Earlier weather cases
4 months

Newest exact grid
2 atmospheres

Absolute ground concentration still requires baseline abundance and a validated column-to-surface mapping.

Notation and units

$c(z)$: mass concentration at altitude z
 c_0 : surface mass concentration
 c and c_0 use the same mass / volume units

z : altitude (m)
 H : assumed concentration scale height (m)
 $\exp$: exponential function

$\hat{c}_0$: estimated surface mass concentration
 $\hat{}$: estimated quantity
 $\Rightarrow$: implication under the assumed profile

Task 1.2 · Spectroscopic extraction

Acquired line positions, strengths, pressure shifts, air broadening and temperature exponents from HITRAN.

Used isotope-specific masses and partition functions. Preserved line-source hashes and failed acquisition requests.

$$\sigma_s(\nu, z) = \sum_{\ell \in s} S_{\ell}(T(z)) V_{\ell}[\nu - \nu_{\ell} - \delta_{\ell}P(z); T(z), P(z)]$$

σ_s is molecular cross section. S_{ℓ} is line strength and V_{ℓ} is the normalized Voigt profile for local temperature and pressure.

The joint receiver now includes five organic gas targets and four interfering gases.

HITRAN strengths already include natural isotope abundance. Applying it twice would understate absorption.

Notation and units

s: species index, ℓ : spectral line index
 ν : spectral frequency, ν_{ℓ} : line centre
z: altitude (m)
 σ_s : molecular absorption cross section (m²)

$S_{\ell}(T)$: integrated line strength at T
 V_{ℓ} : normalized Voigt line profile
 $S_{\ell} \times V_{\ell}$ has cross section units
Use matching spectral units for S_{ℓ} and V_{ℓ}

$T(z)$: atmospheric temperature (K)
 $P(z)$: atmospheric pressure
 δ_{ℓ} : line shift per unit pressure
 $\delta_{\ell} P(z)$: pressure shift in frequency units

Parameter / check	Value
Current target gases	5
Interfering gases	4
Historical isotope tables	34
Historical 60–400 GHz lines	31,474

Task 1.2 · Every target gas and catalog gap

CO, O₃, SO₂ and NO₂ enter as interferents. H₂O and O₂ enter the atmospheric background.

Target	Chemical identity	Current sensing result
H ₂ CO	Formaldehyde, an aldehyde	Weak recall at 1 µg/m ³
CH ₃ OH	Methanol, an alcohol	Weak recall at 1 µg/m ³
CH ₃ CN	Acetonitrile, a nitrile	Strongest low-concentration candidate
CH ₃ Cl	Chloromethane, a halogenated gas	Improves at higher concentration
HCOOH	Formic acid, a carboxylic acid	Largest current 95% concentration limit

Formaldehyde 95% limit
7.846 µg/m³

Acetonitrile 95% limit
0.927 µg/m³

Formic acid 95% limit
30.550 µg/m³

CH₃Br, C₂H₄, CH₃F and CH₃I lack suitable lines in the acquired HITRAN window. This does not prove zero absorption.

Task 1.3 · Voigt profiles and slant propagation

Combined Doppler and pressure broadening at each atmospheric layer.

Integrated along a spherical refracted ray connecting receiver and satellite. Used atmospheric interfaces and numerical quadrature.

$$A_s(\nu) = \frac{10}{\ln 10} \int_{\text{ray}} n_s(z) \sigma_s(\nu, z) ds,$$
$$\frac{P_{\text{out}}}{P_{\text{in}}} = 10^{-A/10}, \quad b = n(r) r \sin \zeta$$

A is power attenuation in dB. b is the spherical ray invariant and ζ is the local zenith angle.

Elevation changes both the optical path and the received SNR.

A secant factor applied after vertical integration cannot represent all refracted path effects.

Notation and units

s: gas species, ν : frequency
A_s and A: power attenuation (dB)
n_s: molecular number density (m⁻³)
 σ_s : molecular cross section (m²)

ds: differential path length along ray (m)
P_in and P_out: input and output power (W)
z: altitude (m)
ln: natural logarithm

b: conserved spherical ray invariant (m)
n(r): refractive index at radial distance r
r: distance from Earth's centre (m)
 ζ : local zenith angle

Parameter / check	Value
Current elevations	30°, 45°, 60°, 90°
Numerical tolerance	0.1% RMS
Gas convergence	0.00381% RMS
Background convergence	0.00200% RMS

Task 1.3 · Numerical convergence and thermal correction

Quantity	Relative RMS difference	Interpretation
Gas integration	0.00381%	Below the declared 0.1% numerical tolerance
PM integration	0.0225%	Below the declared tolerance
Atmospheric background	0.00200%	Below the declared tolerance
Thermal calculation, initial	0.11596%	Failed and triggered refinement
Thermal calculation, refined	0.00319%	Passed without relaxing the tolerance

$$T_{\text{sky}} = \sum_i J_{\nu}(T_i) (1 - e^{-\tau_i}) e^{-\sum_{j<i} \tau_j} + J_{\nu}(T_{\text{space}}) e^{-\sum_i \tau_i}$$

Layer emission contributes receiver noise as well as attenuation.

Lower-atmosphere refinement
100 m steps

Above 20 km
500 m steps

Separate September check
0.05304% RMS

These are numerical checks on the tested grid. They do not measure atmospheric model error.

Notation and units

T_{sky} : equivalent sky noise temperature (K)
 ν : frequency (Hz)
 T_i : physical temperature of layer i (K)

$J_{\nu}(T)$: Planck brightness temperature (K)
 τ_i : dimensionless optical depth of layer i
 i : emitting layer, j : intervening layer

$j < i$: layers between layer i and receiver
 T_{space} : space background temperature (K)
 $\exp(-\tau)$: power transmission through a layer

Task 1.4 · Rayleigh and Mie particle models

Integrated spherical Mie extinction over truncated lognormal particle size distributions and normalized it by dry mass.

Converted aerodynamic size cuts to physical particle dimensions using the assumed density and shape model.

$$\kappa_{\text{ext}} = \frac{\int C_{\text{ext}}(r) p(r) dr}{\int \frac{4\pi}{3} \rho r^3 p(r) dr},$$
$$\kappa_{\text{abs}} \simeq \frac{3k \operatorname{Im}(q)}{\rho}, \quad \kappa_{\text{sca}} \simeq \frac{2k^4 r^3 |q|^2}{\rho}$$

$q = (m^2 - 1)/(m^2 + 2)$. k is wave number, m complex refractive index, ρ density and r sphere radius.

Rayleigh and full Mie agree closely in this particle-size and frequency regime.

Material refractive index, shape, composition and humidity response remain assumptions.

Notation and units

κ_{ext} , κ_{abs} , κ_{sca} : mass coefficients
Extinction, absorption, scattering (m^2/kg)
 $C_{\text{ext}}(r)$: particle extinction cross section (m^2)
 $p(r)$: particle number distribution in radius
 r : physical particle radius (m)

ρ : particle material density (kg/m^3)
 $k = 2\pi/\lambda$: electromagnetic wavenumber (m^{-1})
 λ : wavelength in the surrounding medium (m)
 $q = (m^2 - 1)/(m^2 + 2)$: contrast factor
 m : complex relative refractive index

$\operatorname{Im}(q)$: imaginary part of q
 $|q|^2$: squared complex magnitude
 dr : radius integration element (m)
 $\simeq$: Rayleigh approximation for small particles
The two final terms describe one radius r

Parameter / check	Value
Fine density	1,500 kg/m^3
Coarse density	1,800 kg/m^3
Fine index	$1.5 + 0.01i$
Coarse index	$1.53 + 0.01i$

Task 1.4 · PM definitions and the negative result

Output	Mass represented	Current result
PM1	Below 1 µm aerodynamic diameter	Exploratory extra split fails numerical separation
PM2.5	Fine mass below 2.5 µm	No useful mass retrieval in the tested grid
Coarse PM	Disjoint 2.5–10 µm mass	Nearly collinear with the fine-mode signature
PM10	Fine plus coarse mass	Includes their covariance, no valid 95% limit

$$\text{Var}(\text{PM}_{10}) = \text{Var}(c_{\text{fine}}) + \text{Var}(c_{\text{coarse}}) + 2 \text{Cov}(c_{\text{fine}}, c_{\text{coarse}})$$

The fine model truncates below 0.03 µm. PM10 includes PM2.5, so they are not disjoint particle populations.

Fine Rayleigh/Mie difference
0.00086% RMS

Coarse difference
0.0201% RMS

Three-bin inverse identity error
0.508, rejected

A gain slope removes much of smooth Rayleigh absorption. Shared scattering shapes leave extremely weak size information.

Notation and units

PM₁₀: total fine plus coarse mass (µg/m³)
c_{fine}: modeled PM2.5 mass (µg/m³)
c_{coarse}: modeled 2.5–10 µm mass (µg/m³)

Var: variance of a concentration estimate
Cov: covariance between the two estimates
All variance terms have units (µg/m³)²

Fine and coarse bins are disjoint
PM₁₀ = c_{fine} + c_{coarse}
Their covariance must enter the total error

Task 2.1 · Link budget and sequential receiver

Specified a 550 km downlink, 23 dBm active transmit power, 0.50 / 0.30 m apertures and a nominal 6 dB noise figure.

Replaced the ideal simultaneous broad reference with one RF chain hopping across sixteen 16 MHz blocks within 220–330 GHz.

$$P_r(\nu) = \frac{P_t(\nu)G_t(\nu)G_r(\nu)}{L_{\text{impl}}} \left(\frac{\lambda}{4\pi R} \right)^2 10^{-A(\nu)/10},$$
$$\text{SNR}(\nu) = \frac{P_r(\nu)}{kB [T_{\text{sky}}(\nu) + T_0(F - 1)]}$$

R is range, F the linear noise factor and Limpl the linear implementation loss. Receiver noise figure enters once.

Parameter / check	Value
Power / noise figure	23 dBm / 6 dB
Implementation loss	5 dB
Instantaneous bandwidth	16 MHz
Settling per hop	1 ms

Each block has sixteen 1 MHz tones. CP, pilots, full frames, reference time and retuning all consume the time budget.

23 dBm across the selected bands and 6 dB receiver noise are unverified RF requirements.

Notation and units

ν : frequency (Hz), λ : wavelength (m)
 P_t , P_r : transmit and received power (W)
 G_t , G_r : antenna power gains (linear)
 L_{impl} : implementation loss (linear)
 R : propagation distance (m)

$A(\nu)$: atmospheric attenuation (dB)
SNR: signal to noise power ratio (linear)
 k : Boltzmann constant (J/K)
 B : receiver noise bandwidth (Hz)
 T_{sky} : sky noise temperature (K)

T_0 : receiver noise reference temperature (K)
 F : receiver noise factor (linear)
 $F = 10^{(\text{noise figure in dB} / 10)}$
 $T_0(F - 1)$: receiver noise temperature (K)
Gains and losses must be converted from dB

Task 2.1 · Resources and communication cost

Quantity	Current result
Total reference + sample time	20 s, split equally
Actual frame transmission / retuning	19.72 s / 0.032 s
Instantaneous bandwidth / RF chains	16 MHz / one
Selected schedule Gaussian rate	109.52 Mbit/s
Best tested fixed block Gaussian rate	128.31 Mbit/s
Rate loss against that fixed block	14.64%

$$\bar{R} = \sum_b \frac{t_b}{t_{\text{total}}} \sum_{k \in b} \Delta f \log_2(1 + \text{SNR}_k)$$

The comparison uses equal instantaneous bandwidth, power and CP/pilot overhead. Both reference and sample carry payload.

Selected center range
224–318.67 GHz

Pilot / frame symbols
30 / 10,000

CP / useful symbol
1 / 16

Zero architecture cost is not established. Gaussian rate is different from coded QPSK throughput.

Notation and units

$\bar{R}$: average Gaussian information rate (bit/s)
 b : selected frequency block index
 k : tone index within block b

t_b : transmission time in block b (s)
 t_{total} : total acquisition time (s)
 t_b / t_{total} : block's time fraction

Δf : bandwidth per tone (Hz)
 SNR_k : tone signal to noise ratio (linear)
 $\log_2$: base two logarithm

Task 2.2 · CSI and raw waveform simulation

Generated complex line-of-sight observations with frequency-dependent attenuation and additive complex Gaussian noise.

Added raw IFFT/CP frames, timing acquisition, Doppler correction, pilot tracking and coded packet checks.

$$y_k = h_k x_k + w_k, \quad w_k \sim \mathcal{CN}(0, V_k),$$
$$x[n] = \text{IFFT}\{X_k\}$$

Separate raw frames test modem behavior. Large recall experiments draw joint sample moments and apply nonlinear inversion.

The simulation connects physical channel attenuation to a bounded moving receiver.

The large recall study does not generate every raw waveform sample for every Monte Carlo trial.

Notation and units

k: tone index, n: time sample index
x_k: transmitted tone symbol
y_k: received tone symbol
h_k: complex channel coefficient

w_k: additive complex receiver noise
V_k: noise variance in symbol power units
CN(0, V_k): circular complex Gaussian noise
Its real / imaginary variances are V_k / 2

X_k: frequency domain OFDM symbols
x[n]: transmitted time domain samples
IFFT: inverse fast Fourier transform
h_k scales amplitude and rotates phase

Parameter / check	Value
Blocks / tones per block	16 / 16
Frame length	10,000 symbols
Pilots per frame	30
Acquisition CFO stress	100 kHz

Task 2.2 · What the waveform controls establish

Control	Retained result	Boundary
Selected-frequency coverage	16 fresh raw frames	One frequency spot check per selected block
Timing acquisition	16 / 16 correct	Bounded integer timing model
Passive sensing	Identical decoded outputs in every case	Comparison uses the same scheduled modem
Coded information errors	1,391 errors	Across 2,826,240 information bits
Full mission dynamics	Still open	Fractional timing, clock drift and time dilation

$$f_D \simeq \frac{v_r}{c} f_c$$

Narrowband Doppler correction addresses carrier shift. Wideband time scaling requires additional receiver modeling.

Information bits
2,826,240

Information bit errors
1,391

Raw coded bit error rate
0.0492%

Oscillator noise, pointing, multipath and realistic reference availability remain open.

Notation and units

f_D : Doppler frequency shift (Hz)
 f_c : carrier frequency (Hz)

v_r : signed radial relative velocity (m/s)
 c : speed of light (m/s)

$\simeq$: first order narrowband approximation
The shift sign follows the velocity convention

Task 3.1 · Joint spectral isolation

Fitted all five gas signatures and two PM modes simultaneously.

Whitened the observation covariance and projected out gain offset/slope, atmospheric background and four interfering gases.

$$\mathbf{a} = \mathbf{D}\mathbf{c} + \mathbf{N}\boldsymbol{\beta} + \mathbf{e}, \quad \mathbf{C} = \text{Cov}(\mathbf{e}),$$
$$\mathbf{P} = \mathbf{I} - \mathbf{Q}_N\mathbf{Q}_N^T, \quad \hat{\mathbf{c}} = \mathbf{H}\mathbf{a}$$

D contains target signatures. N contains nuisance signatures. QN spans the whitened nuisance space. An SVD constructs H.

Numerical gates check scaled $\mathbf{H}\mathbf{D} \approx \mathbf{I}$ and nuisance rejection $\mathbf{H}\mathbf{N} \approx \mathbf{0}$.

No algorithm can separate an arbitrary gain error that occupies the same spectral direction as the pollutant.

Notation and units

a: attenuation vector (dB), e: error (dB)
D: target response matrix
c, $\hat{c}$: true / estimated concentrations ($\mu\text{g}/\text{m}^3$)
N β : nuisance attenuation contribution (dB)

C = Cov(e): observation covariance (dB^2)
N: nuisance response matrix
 β : nuisance coefficients
Q_N: orthonormal whitened nuisance basis

P: projector away from nuisance space
I: identity matrix, T: matrix transpose
H: linear concentration retrieval operator
Bold symbols denote vectors or matrices

Parameter / check	Value
Unknown target masses	7
Reported outputs	8 with PM10
Selected frequencies	256 tones
Scaled identity tolerance	10^{-5}

Task 3.1 · Frequency selection improves five-gas inference

A deterministic coordinate-exchange screen selected sixteen blocks using the standard 45° design case.

The selection optimized the worst relative gas uncertainty while retaining PM and nuisance parameters. Exact tone spectra followed the coarse screen.

CH ₃ CN at 1 µg/m ³	Uniform bands	Selected bands
Simulated recall	57.40%	97.67%
Total acquisition time	20 s	20 s
Assumed residual standard deviation	0.0001 dB	0.0001 dB

All five exact VOC information limits improve relative to the uniform design.

Deterministic starts

3

Selected blocks

16

Improved exact gas limits

5 / 5

January weather and response draws did not choose bands. Global optimality and unknown-weather robustness remain unproved.

Task 3.2 · Payload moments estimate signal and noise

Used the existing constant-modulus QPSK payload instead of discarding most samples and retaining only 30 pilots per 10,000 symbols.

Applied the established M2M4 estimator with a finite-sample correction, then compared sample and reference powers.

$$M_2 = S + V, \quad M_4 = S^2 + 4SV + 2V^2,$$
$$S = \sqrt{2M_2^2 - M_4}, \quad A = -10 \log_{10} \left(\frac{S_{\text{sample}}}{S_{\text{reference}}} \right)$$

S is received signal power. V is noise power. The moments separate signal and noise under the QPSK and Gaussian-noise model.

More existing symbols contribute information without adding sensing transmissions.

Arbitrary QAM, nonlinear distortion or severe intercarrier interference invalidate these simple moments.

Notation and units

M_2 : second moment $E[|y|^2]$
 M_4 : fourth moment $E[|y|^4]$
 y : received complex symbol
 E : expectation, $|y|$: complex magnitude

S: received signal power
V: receiver noise power
 M_2 , S, V share power units
 M_4 has squared power units

A: differential attenuation (dB)
 S_{sample} : signal power in sample observation
 $S_{\text{reference}}$: signal power in reference
 $\log_{10}$: base ten logarithm

Parameter / check	Value
Pilot fraction per frame	0.3%
Payload fraction per frame	99.7%
Signal moments	Second and fourth
Modulation tested	QPSK

Task 3.2 · Why payload reuse mattered

Earlier ideal broad reference, three target gases and zero persistent calibration residual

Historical control	Pilots	Payload M2M4	Reference assumption
CH ₃ CN, 1 µg/m ³ , 10 s	0.460% recall	99.955% recall	Exactly known reference
CH ₃ CN, 1 µg/m ³ , 20 s total	0.380% recall	93.800% recall	10 s independent reference + 10 s sample

Payload moments recover information that sparse pilot-only sensing discards.

Pilot samples per frame

30

Total symbols per frame

10,000

Old physical architecture

Ideal simultaneous broad bands

The ideal broad architecture and zero residual assumption prevent direct transfer to deployed hardware.

Task 3.2 · Detection gains are species dependent

20 s total, assumed 0.0001 dB residual, selected bands and standard atmosphere at 45°

Gas	Recall at 1 (%)	Misses at 1 (%)	c95 $\mu\text{g}/\text{m}^3$	Recall at c95 (%)	95% CI at c95 (%)	RMSE at c95 (%)
Formaldehyde	0.77	99.23	7.846	95.12	94.68–95.53	21.14
Methanol	0.75	99.25	8.838	95.12	94.68–95.53	21.24
Acetonitrile	97.67	2.33	0.927	95.12	94.68–95.53	21.49
Chloromethane	3.58	96.42	3.918	94.72	94.26–95.15	21.64
Formic acid	0.16	99.84	30.550	94.85	94.40–95.28	21.55

Acetonitrile reaches 97.67% recall at 1 $\mu\text{g}/\text{m}^3$. Other gases require higher concentrations.

Positive / null trials
10,000 / 10,000

Family false alarms
0.79%

Relative RMSE at 95% points
Approximately 21%

These are simulated detection frequencies, not measured field performance or concentration accuracy.

Task 4.1 · Lower bounds with nuisance parameters

Derived full QPSK and magnitude likelihood information with unknown signal power, noise power and finite reference uncertainty.

Profiled nuisance parameters before comparing the attainable concentration covariance with the estimator variance.

$$J_{ab} = \mathbb{E} \left[\frac{\partial \log p(y; \boldsymbol{\theta})}{\partial \theta_a} \frac{\partial \log p(y; \boldsymbol{\theta})}{\partial \theta_b} \right],$$

$$\mathbf{J}_{\text{eff}} = \mathbf{J}_{cc} - \mathbf{J}_{c\eta} \mathbf{J}_{\eta\eta}^{-1} \mathbf{J}_{\eta c}, \quad \text{Cov}(\hat{\mathbf{c}}) \succeq \mathbf{J}_{\text{eff}}^{-1}$$

c denotes target concentration and η all nuisance parameters. The Schur complement accounts for their uncertainty.

Parameter / check	Value
Likelihoods	QPSK and magnitude
Unknown noise	Included
Finite reference	Included
Historical efficiency	95.89–97.61%

The bound identifies when replacing the estimator offers little remaining gain.

The bound is local and conditional on the likelihood, covariance and physical signatures.

Notation and units

y: observed signal data
 θ: complete unknown parameter vector
 p(y;θ): likelihood of the observations
 E: expectation under the assumed model

J_ab: Fisher information entry for θ_a, θ_b
 c: target concentration vector (μg/m³)
 η: nuisance parameter vector
 J_cc, J_cη, J_ηη: Fisher matrix blocks

J_eff: information after nuisance profiling
 ĉ: estimated concentration vector
 Cov(ĉ): estimator covariance, units (μg/m³)²
 -1: matrix inverse, ≥: covariance lower bound

Task 4.1 · M2M4 is already close to the bound

Historical broad reference, three target gases, 20 s total and zero extra calibration residual

Gas	CRLB / M2M4 variance	Historical 95% limit µg/m³
Formaldehyde	95.89%	4.456
Methanol	97.61%	10.307
Acetonitrile	96.49%	1.023

$$\text{Efficiency} = \frac{\text{variance lower bound}}{\text{estimator variance}}$$

Values near 100% leave little improvement available from estimator replacement in this control.

The next substantial improvement came from frequency information and design.

Historical output family
6

Acquisition split
10 s + 10 s

Calibration benchmark
Zero extra residual

These historical bounds use a different architecture and target family from the current five-gas receiver.

Notation and units

Efficiency: lower bound / estimator variance
Displayed percentage = 100 × this ratio

Variance lower bound: conditional CRLB
CRLB: Cramér–Rao lower bound

Estimator variance: variance of M2M4 output
Both variances use the same concentration units²

Task 4.2 · Time and calibration jointly set uncertainty

Recomputed the link, absorption, emission and target fit when weather, elevation, bands or receiver noise change.

The latest common grid contains every VOC, PM2.5, coarse PM and PM10 in every operating condition.

$$\mathbf{C}(t, \sigma_{\text{cal}}) = \text{diag}[\mathbf{V}_{\text{sample}}(t/2) + \mathbf{V}_{\text{reference}}(t/2)] + \sigma_{\text{cal}}^2 \mathbf{K},$$
$$K_{ij} = \exp\left(-\frac{|\nu_i - \nu_j|}{10 \text{ GHz}}\right)$$

Thermal variance decreases with retained symbols. The additional calibration residual persists through the acquisition.

More time helps until persistent calibration error dominates.

The 10 GHz correlation length and each calibration standard deviation remain unmeasured assumptions.

Notation and units

C: attenuation covariance matrix (dB²)
t: total reference plus sample time (s)
t / 2: time assigned to each observation
V_sample, V_reference: variance vectors (dB²)

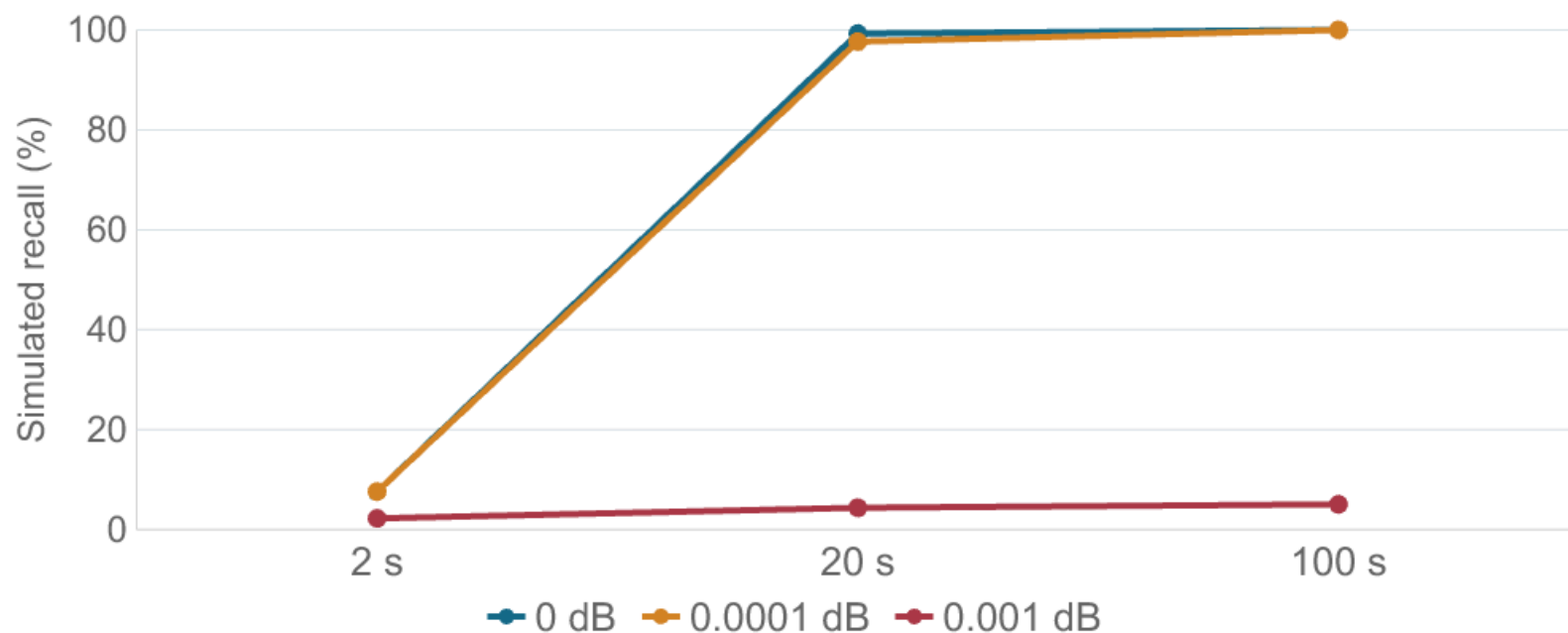
diag: diagonal matrix built from a vector
 σ_{cal} : persistent residual standard deviation (dB)
K: dimensionless spectral correlation matrix
K_ij: correlation of frequency bins i and j

ν_i, ν_j : bin frequencies in matching units
10 GHz: assumed spectral correlation length
exp: exponential function
 σ_{cal}^2 K: persistent covariance contribution

Parameter / check	Value
Time grid	2 / 20 / 100 s
Residual grid	0 / 0.0001 / 0.001 dB
Noise-figure grid	6 / 17 dB
Calibration draws	Once per acquisition

Task 4.2 · Longer time cannot remove calibration error

Acetonitrile at 1 µg/m³, selected bands, standard atmosphere at 45°



Parameter / check	Value
20 s, 0 dB	99.28% recall
20 s, 0.0001 dB	97.67% recall
20 s, 0.001 dB	4.36% recall
100 s, 0.001 dB	5.05% recall

At 0.001 dB, recall is only 4.36% at 20 s and 5.05% at 100 s.

Zero and nonzero residuals are sensitivity assumptions. No receiver stability measurement establishes them.

Task 4.2 · PM shares the full global comparison

Variable	Compared settings
Frequency plan	Uniform and selected sixteen-block schedules
Atmosphere	Standard and January sounding
Elevation	30°, 45°, 60°, 90°
Reference + sample time	2 s, 20 s, 100 s
Assumed calibration residual	0, 0.0001, 0.001 dB
Receiver noise figure	6 dB and 17 dB

288 conditions × 8 outputs = 2,304 rows. Each output has 270 computed and 18 rejected conditions.

Common result rows

2,304

Rejected settings

Standard, 30°, NF 17 dB

PM predicted recall

Approximately 0.125%

No valid 95% response limit for PM2.5 or PM10 anywhere in this grid.

Task 4.3 · Detection limits and percentage definitions

Fixed a 1% family false-alarm budget across eight outputs before observing outcomes.

Solved the concentration for nominal 95% power using positive-signal variance, then tested it with fresh simulated responses.

$$z = \Phi^{-1}\left(1 - \frac{0.01}{8}\right), \quad \hat{c}_j > z s_{\text{null}},$$
$$c_{95} = z s_{\text{null}} + 1.64485 s_{\text{positive}}(c_{95})$$

Parameter / check	Value
One-sided target α	$0.01 / 8 = 0.00125$
Null threshold z	Approximately 3.023
Positive power target	95%
Absorption cap	1 dB

Positive absorption changes SNR and variance. The solution stays within at most 1 dB added absorption.

Reports include low recalls, missed detections, exact binomial intervals and concentration RMSE.

95% detection power, a 95% confidence interval and 5% concentration error describe different quantities.

Notation and units

Φ^{-1} : inverse standard normal CDF
 z : dimensionless null detection threshold
0.01: family false alarm probability budget
8: number of reported outputs

$\hat{c}_j$: estimate for target j ($\mu\text{g}/\text{m}^3$)
 s_{null} : standard deviation under the null
 $s_{\text{positive}}(c)$: standard deviation at mass c
Both standard deviations use $\mu\text{g}/\text{m}^3$

c_{95} : concentration for 95% recall ($\mu\text{g}/\text{m}^3$)
1.64485: $\Phi^{-1}(0.95)$, a normal quantile
>: target detection decision
The variance depends on positive concentration

Task 4.3 · Every 95% limit needs time and calibration

Nominal 95% response concentration in $\mu\text{g}/\text{m}^3$, selected bands, standard atmosphere at 45°

Total s	Residual dB	H ₂ CO	CH ₃ OH	CH ₃ CN	CH ₃ Cl	HCOOH
2	0	25.778	27.695	2.917	12.206	102.781
2	0.0001	25.862	27.911	2.938	12.309	102.894
2	0.001	33.018	44.262	4.550	19.693	113.332
20	0	7.568	8.131	0.856	3.584	30.171
20	0.0001	7.846	8.838	0.927	3.918	30.550
20	0.001	21.913	35.377	3.557	15.567	56.267
100	0	3.361	3.612	0.380	1.592	13.400
100	0.0001	3.945	4.999	0.517	2.226	14.228
100	0.001	20.826	34.603	3.468	15.192	49.310

CH₃CN, 20 s / 0.0001 dB

0.927 $\mu\text{g}/\text{m}^3$

CH₃CN, 100 s / 0.001 dB

3.468 $\mu\text{g}/\text{m}^3$

Valid PM limits in nine cases

0

PM_{2.5}, coarse PM and PM₁₀ have no valid 95% response limit in these nine conditions.

Task 4.3 · Environmental interpretation remains open

Converted gas mass to surface-equivalent mole fraction using pressure, temperature and molar mass.

Observable: short slant-path enhancement relative to a reference. Ambient guidance has distinct location and averaging requirements.

PM reference masses are 15 and 45 µg/m³. These tests do not establish compliance.

$$\chi_{\text{ppm}} = \frac{c_{\mu\text{g}/\text{m}^3} R T}{p M_{\text{g}/\text{mol}}}$$

The numeric conversion assumes the displayed mass and molar-mass units. PM has mass units and no gas ppm conversion.

Parameter / check	Value
H ₂ CO current limit ppm	0.006179
CH ₃ CN current limit ppm	0.000534
Limit conditions	20 s / 0.0001 dB
Ambient PM test masses	15 and 45 µg/m³

The study reports conditional floors and an explicit negative / unverified standards assessment.

Unknown baseline abundance, vertical profiles, PM information and independent truth prevent a positive compliance claim.

Notation and units

χ_ppm: gas mole fraction in ppm
c: gas mass concentration in µg/m³
M: gas molar mass in g/mol

R: ideal gas constant, 8.314 J/(mol K)
T: local absolute temperature (K)
p: local atmospheric pressure (Pa)

1 ppm = 10⁻⁶ mol/mol
The stated mass units cancel the 10⁶ factor
This gas conversion does not apply to PM

Calibration is the critical engineering requirement

Zero calibration error means zero additional differential residual after modeled corrections.
Thermal noise and reference uncertainty still remain.

At 20 s, the local CH₃CN requirement for 95% power at 1 µg/m³ is approximately 0.000146 dB residual standard deviation under the nominal covariance.

$$|b_i| \leq \varepsilon \implies B_j = \varepsilon \|\mathbf{H}_j\|_1,$$
$$c_{95,\text{protected}} \simeq (z + 1.64485)s_j + 2B_j$$

Parameter / check	Value
Required local σ at 20 s	About 0.000146 dB
Nominal random σ	0.0001 dB
Power change for 0.0001 dB	About 0.002303%
Field calibration records	None

A persistent signed bias can move both the null threshold and the positive response in the adverse direction.

Calibration requirements are now quantitative and retained with every result.

A small random standard deviation does not guarantee resistance to arbitrary drift or unknown hop gains.

Notation and units

b_i : persistent bias in attenuation bin i (dB)
 ε : upper bound on each $|b_i|$ (dB)
 i : spectral bin, j : retrieved target

$\mathbf{H}_j$: row j of the concentration operator
 $\|\mathbf{H}_j\|_1$: sum of absolute row coefficients
 B_j : resulting concentration bias bound (µg/m³)

s_j : concentration standard deviation (µg/m³)
 z : null threshold, 1.64485: 95% normal quantile
 $c_{95,\text{protected}}$: bias protected limit (µg/m³)
 $\simeq$: local approximation with constant variance

Failures that constrain the claimed achievement

Failure or stress	Finding	Implication
PM mass and extra size split	Negligible useful information / unstable inverse	A successful optimizer cannot repair missing information
Unknown relative hop gains	Target separation fails	Requires calibrated relative spectral response
Weather mismatch	Frozen standard operator acquires bias	Matched-weather success is insufficient
Unamplified RF converter	Fails usable-SNR gate	Nominal power requires a complete RF chain
Modulation / receiver mismatch	Simple payload moments may become invalid	Receiver assumptions need dedicated validation

Numerically rejected global cases

18 / 288

Useful PM limits

0 / 270 valid cases

Gaussian architecture rate loss

14.64%

The intended general atmospheric VOC/PM instrument remains an open research outcome.

Remaining problems and evidence needed

Open problem	Evidence required to close it
RF hardware and calibration	Measured power, noise, settling, spectral covariance and drift at selected tones
PM mass / composition	Paired optical properties, humidity, particle size and independent mass truth
Weather and concentration profiles	Independent profiles, column information and reference abundance
Full receiver dynamics	Continuous time-scaled channel with fractional timing, oscillator and pointing errors
Independent validation	Separate calibration and test campaigns with paired radio / concentration data
Deployment and environmental use	Applicable averaging domain, spectrum coordination and realistic operational coverage

Nominal RF target
23 dBm, NF 6 dB

Current calibration assumption
0.0001 dB

Current physical truth
No paired radio/VOC/PM test

Proposed next experiments

Priority	Experiment	Decision it enables
1	Repeated blank / reference acquisitions at selected tones	Can the required differential stability be achieved?
2	Controlled gas path with independent concentration truth	Does the predicted VOC recall survive real model error?
3	PM cell with known mass, size and humidity	Does radio information separate mass from composition?
4	Full receiver and reference schedule under motion	Can the sensing result survive operational timing?

Freeze calibration and model choices before evaluating an independent test campaign.

Calibration acceptance question
 $\sigma \leq \text{about } 0.000146 \text{ dB at } 20 \text{ s?}$

Frozen test target
 $1 \mu\text{g}/\text{m}^3 \text{ CH}_3\text{CN}$

Independent test measurements
Still required

Additional Monte Carlo trials improve numerical precision but cannot replace missing measurements.

What we can report to the supervisor

The original eleven modeling and simulation tasks now have implementations with traceable outputs.

Payload reuse and frequency selection improve conditional VOC detection. Acetonitrile reaches 97.67% recall at 1 $\mu\text{g}/\text{m}^3$ with 20 s and assumed 0.0001 dB residual.

The analysis explains why PM remains weak and why more averaging cannot overcome persistent calibration error.

The next milestone is independent physical validation of the selected receiver and its concentration estimates.

The main achievement is a reproducible feasibility assessment and a stronger, explicitly conditional VOC design.

Useful joint PM retrieval, zero communication architecture cost and validated field performance remain unachieved.

Parameter / check	Value
Task implementations	11 / 11
Verified research checks	25,172
Conditional CH ₃ CN recall	97.67%
Validated field recall	Unavailable

Appendix · Global VOC and PM outcomes

Envelopes over 270 valid operating conditions per output, with 18 additional rejected conditions

Output	Reference $\mu\text{g}/\text{m}^3$	Predicted recall range	Valid 95% limits / 270
H ₂ CO	1	0.128–89.043%	270
CH ₃ OH	1	0.130–72.886%	270
CH ₃ CN	1	0.282–100.000%	270
CH ₃ Cl	1	0.139–100.000%	270
HCOOH	1	0.125–12.651%	269
PM _{2.5}	15	0.125–0.125%	0
PM _{coarse}	45	0.125–0.125%	0
PM ₁₀	45	0.125–0.125%	0

Grid
2 plans × 2 weather × 4 elevations

Other axes
3 times × 3 residuals × 2 noise figures

Mass assumptions
1 / 15 / 45 $\mu\text{g}/\text{m}^3$ (see table)

Different reference concentrations and operating conditions prevent a direct same-condition species ranking.

Appendix · Low PM recalls remain visible

Simulated recall, selected bands, standard atmosphere at 45°, same joint five-gas fit

Total s	Residual dB	PM2.5 at 15 µg/m³	Coarse at 45 µg/m³	PM10 at 45 µg/m³
2	0	0.14%	0.14%	0.12%
2	0.0001	0.10%	0.10%	0.06%
2	0.001	0.09%	0.11%	0.14%
20	0	0.11%	0.15%	0.16%
20	0.0001	0.15%	0.07%	0.14%
20	0.001	0.15%	0.19%	0.09%
100	0	0.15%	0.12%	0.10%
100	0.0001	0.08%	0.13%	0.07%
100	0.001	0.07%	0.10%	0.13%

PM10 positive mixture
Fine 49/90, coarse 41/90

Null target threshold
0.125% theoretical marginal rate

Monte Carlo class size
10,000

These recalls are near the false-alarm scale. No cell establishes useful PM detection.

Appendix · Reproducible evidence map

Research snapshot 461a3ba, 27 September 2026

Research material	Repository location
Original task definition	I2R_proposal_THz_ISAC (1).pdf
Current results and global comparison	docs/51_joint_design_time_calibration_results.md
Species, equations and metric definitions	docs/50_species_methods_and_percentage_guide.md
Complete open-issue ledger	docs/49_completion_audit_and_fixes.md
Critical calibration record	docs/46_critical_calibration_assumption.md
Data, trials, hashes and validation	results/joint_receiver_revision/
Current manuscript	output/pdf/thz_isac_pollutant_sensing_ieee.pdf

Retained metric rows

480

Global rows

2,304

Passing research tests

235

Appendix · Physical and methodological sources

HITRAN2024 and HAPI: line data, isotope handling and spectroscopic calculations.

ITU-R P.835 and P.676: reference atmosphere, oxygen/water absorption and atmospheric background.

NOAA IGRA: independent measured weather soundings.

Aliaga et al. (2024) and Yang et al. (2024): satellite differential absorption and gas/particle propagation context.

Xu, Fu and Kim (2025): using pilots and payloads for sensing. Established M2M4 literature supplies the signal/noise estimator.

Parameter / check	Value
Original tasks	11
Current source snapshot	461a3ba
Additional organic catalog gaps	4
Paired field validation	Open

Published models motivate and parameterize the study. They do not validate this receiver's field performance.